# Learning to Monitor the Reliability Horizon of Imagined Futures


## Abstract

An agent that plans by imagining must decide how far to trust what it imagines. We study whether a world model can learn a *monitor* of that trust: a predictor of the *reliability horizon* $H^*$, the number of consecutive imagined steps whose predicted state matches the environment exactly. In partially observed gridworlds, a learned monitor predicts $H^*$ with Spearman $\rho = 0.573 \pm 0.011$ and improves on an *optimistically fitted* monotone single-scalar ceiling by 1.04 MAE on average, with per-seed bootstrap confidence intervals excluding zero. The monitor requires information from the imagined rollout: restricted to time-$t$ features it fails to beat that ceiling on any seed. Under distribution shift the monitor fails *anti-conservatively*, predicting horizons near 20 where the true horizon is 4–8; it is most confident where the world model is least reliable. Two repairs we tested, rank-normalising the monitor's inputs and gating on latent novelty, do not recover it: the uncertainty signals retain aggregate separation between distributions while losing within-distribution rank-informativeness. Refitting the monitor with a single shifted configuration in its training set moves a never-seen configuration from MAE 16.77 / $\rho$ 0.135 to MAE 3.58 / $\rho$ 0.766, and a size-matched control indicates this is not explained by sample count alone. The restoration is selective: shifts that alter geometry transfer strongly to other geometric shifts ($\rho$ 0.73–0.86) but weakly to noise-driven ones ($\rho \approx 0.31$). The core result also appears in a world model with no recurrence and no autoregressive latent feedback, though monitors transfer only partially across architectures. These results suggest that simulation-based reliability monitoring is measurable and not architecture-specific, but that its calibration is learned from the distribution of failures the world model has experienced rather than being an invariant property of uncertainty.


## 1 Introduction

World models are used to plan by imagining: an agent rolls its learned dynamics forward under candidate actions and acts on the result [Ha and Schmidhuber, 2018, Hafner et al., 2019, 2020]. The question usually asked of such a model is whether it predicts the future accurately. The question an acting agent actually needs answered is different and sharper: *for how many steps does this particular imagined rollout remain worth acting on?* Too short a horizon wastes the model; too long, and the agent plans against a hallucination.

The standard answer thresholds an uncertainty signal — roll forward, stop when ensemble disagreement or predictive entropy crosses a bound [Janner et al., 2019, Chua et al., 2018]. That recipe rests on a premise which prior work supports: uncertainty correlates with realised rollout error. We call this **Claim A**. It licenses ranking states by riskiness, and nothing more.

We study a stronger statement. **Claim B**: an agent can learn a *monitor* that predicts the horizon itself, per state, as a quantity. The distinction matters because a threshold rule is only ever as good as the monotone map from uncertainty to trust. If that map exhausts the available signal, then nothing beyond Claim A has been demonstrated, however elaborate the machinery.

Our central methodological device is therefore a control rather than a baseline. For each uncertainty scalar we fit an isotonic regression from that scalar to $H^*$ *on the evaluation data itself*. This is deliberately optimistic — it is the best attainable performance of *any* monotone function of that signal, with no generalisation gap. Beating it is the minimum evidence that Claim B says something Claim A does not.

Having established that the monitor works in distribution, we ask when it stops working, whether simple corrections recover it, and what kind of experience does. The negative results carry most of the content: our aim is a characterisation of when simulation-based reliability monitoring works and fails, not a new uncertainty metric.

**Contributions.**

1. An operational notion of the imagined reliability horizon with an exactly checkable target, and evidence that a learned rollout monitor predicts it beyond an optimistic single-scalar uncertainty ceiling (§6.1).
2. Evidence that information from the imagined rollout is necessary: a monitor restricted to time-$t$ features does not beat that ceiling on any seed (§6.2).
3. A failure mode under distribution shift in which the monitor becomes *anti-conservative*, which we trace to loss of feature rank-informativeness rather than of scale (§6.3, §7).
4. Evidence that exposure to a shifted distribution can substantially restore transfer, that the effect is not explained by sample count alone, and that it is selective across shift types and across world-model architectures (§6.5–§6.7).

## 2 Related Work

**World models and latent imagination.** Latent dynamics models trained from pixels or symbols underpin modern model-based control [Ha and Schmidhuber, 2018, Hafner et al., 2019, 2020, 2021, 2023]. We adopt the same recurrent state-space structure, but our object of study is not control performance; it is the horizon over which the imagined rollout is exact.

**Uncertainty in learned dynamics.** Deep ensembles and their disagreement are the standard epistemic signal [Lakshminarayanan et al., 2017, Chua et al., 2018], alongside dropout- and distribution-based estimates [Gal and Ghahramani, 2016, Kendall and Gal, 2017]. Recent work propagates epistemic uncertainty through long autoregressive rollouts and uses it to make offline model-based RL work on physical robots [Li et al., 2025]. These establish Claim A. We treat the best monotone map from such a scalar to the horizon as a control to be beaten, not as a method to be improved.

**Adaptive rollout length.** Model-based RL truncates imagined rollouts when uncertainty exceeds a threshold [Janner et al., 2019], and rollout length can be tuned online by treating it as a meta-level decision problem [Bhatia et al., 2022]. These are *online* rules that observe the rollout, or the training run, as it unfolds; we include threshold truncation as a baseline. Related adaptive machinery appears in latent visual MPC: ELVIS gates a time-varying $\lambda$ — trading bootstrapping against look-ahead — using an upper-confidence-bound score from an ensemble of latent critics [Du et al., 2026]. That adapts the *value estimator*, not a predicted horizon. Our monitor differs from all of these in predicting a horizon as a quantity and being evaluated against ground truth rather than by downstream return.

**Horizon certificates.** Two recent works certify how long a learned model can be trusted. Wang [2026a] derives predictability certificates for equivariant latent world models from the finite-time Lyapunov spectrum, with expanding channels giving logarithmic horizons and contracting channels bounded error floors. Wang [2026b] calibrates trust-horizon curves by split conformal prediction and transports them across a symmetry orbit, reporting a median certified-to-measured horizon ratio of 0.67 and zero anti-conservative violations across their stable audits. Both require known group symmetry and produce certificates that are orbit-level and worst-case; ours is a learned per-state estimate with no symmetry assumption and, notably, *no* such conservatism guarantee — the anti-conservative failure we document in §6.3 is precisely what a conformal construction is designed to exclude.

**Self-estimated rollout error.** Scheinker [2026] trains a bidirectional latent diffusion model and uses the discrepancy of a forward-then-backward rollout as a self-supervised proxy for rollout error, reporting Spearman correlations of 0.91–0.98 against realised error on magnetohydrodynamics, astrophysical mixing layers and face video. That method requires a model trained to run in both directions and an extra backward pass, and its domains contain no acting agent; our §6.2 result — that a forward rollout is necessary — puts us on the same side of the a-priori boundary, at lower cost and without invertibility. Song and Cai [2026] instead explains rollout error post hoc in graph world models, splitting long-horizon error into a topology factor governed by the graph spectral radius and a model factor; that analysis is model-level rather than per-state.

**Failure prediction and data coverage.** Berger et al. [2026] report a negative result closely related to ours: in Dreamer-style RSSM latent rollouts, ensemble disagreement captures local epistemic uncertainty but does not reflect error accumulated over long rollouts, because rollouts are pulled toward well-modelled attractor regions where uncertainty declines even as deviation grows. Hansen and Wang [2026] argue that hallucination in large visual world models is predictable from coverage-related signals and preventable by coverage-aware data collection, concluding it is "inherently a data coverage issue". Our §6.5 result is convergent evidence from a different direction: what fails under shift is repaired by exposure to the right kind of failure, not by better use of the uncertainty signal already available.

**Calibration, selective prediction, and shift.** Calibration of modern networks [Guo et al., 2017], selective prediction with a reject option [Geifman and El-Yaniv, 2017], and distribution-free coverage via conformal methods [Angelopoulos and Bates, 2021] all target when a prediction should be trusted. Our setting differs in that the quantity to be trusted is a *depth* rather than a label, and in that the failure we document is a loss of ranking rather than of coverage. That uncertainty estimates degrade under shift is well documented [Ovadia et al., 2019, Hendrycks and Gimpel, 2017]; we add that the degradation here inverts the safety-relevant direction.

**Positioning.** To our knowledge the combination we study — a learned, per-state horizon monitor, evaluated against an optimistic monotone control, and characterised across distribution shifts and world-model architectures — has not been reported.

## 3 Problem Formulation

**Setting.** An agent observes $o_t$ and acts with $a_t$. A world model with parameters $\theta$ maintains a belief $(h_t, z_t)$ and predicts forward without further observations. Given

an action sequence $a_{t:t+H-1}$ known at planning time, the model produces an *imagined rollout*

$$\begin{aligned}\hat{\tau}_{t:t+H} &= \big(\hat{s}_{t+1}, \hat{s}_{t+2}, \ldots, \hat{s}_{t+H}\big),\\ \hat{s}_{t+k} &= \mathrm{dec}_\theta(h_{t+k}, z_{t+k}).\end{aligned} \tag{1}$$

**Reliability horizon.** With $E_H = d(\hat{s}_{t+H}, s_{t+H})$ the discrepancy against the realised state and a tolerance $\varepsilon$, we define

$$H^*(s_t) \;=\; \max\big\{\, H \le H_{\max} \;:\; E_h < \varepsilon \;\; \forall\, h \in 1..H \,\big\}. \tag{2}$$

Equation (2) is a *first-failure* definition: once imagination diverges it is operationally unusable even if it later resynchronises by coincidence. We set $H_{\max} = 30$.

**Instantiation.** Observations are $5 \times 5$ egocentric symbolic views, so grid size never enters the observation. The model predicts size-agnostic state *factors* — direction dir, a binary moved indicating whether the transition changed the agent's cell, plus carrying and door state — and position is reconstructed by integration,

$$\hat{p}_{t+H} - p_t \;=\; \sum_{k=1}^{H} \mathrm{moved}_{t+k} \cdot u(\mathrm{dir}_{t+k-1}), \tag{3}$$

with $u(\cdot)$ the unit vector of a direction. $E_H$ is the Manhattan error of (3) and $\varepsilon = 1$, so $E_H < \varepsilon$ means the model places the agent in *exactly* the correct cell. We verify (3) exactly on the true factors for every configuration before modelling.

This makes the object of study concrete. The load-bearing factor is moved — whether the cell ahead is free — which is precisely what partial observability hides. Imagination stays exact roughly as long as the planned path remains inside the region the agent has already seen.

**The monitor.** Let $x_t$ be a feature vector computed at time $t$, grouped as: *a-priori* features (belief state, one-step uncertainty, statistics of the current observation), *rollout* features (per-step ensemble disagreement, predictive entropy, spread of imagined displacement across ensemble members), and *action* features (the planned sequence). The monitor

$$\hat{H}_t = g_\phi(x_t), \qquad g_\phi : x_t \mapsto \Delta^{H_{\max}+1} \tag{4}$$

outputs a distribution over $H^* \in \{0, \ldots, H_{\max}\}$, giving both a point estimate (its mean) and $P(H^* \ge h)$ for calibration. Crucially, $x_t$ may not depend on any observation, state, or error at time $> t$; we verify this by corrupting every future field and checking that all features are bit-identical while $H^*$ changes.

**The Claim-A ceiling.** For a scalar uncertainty signal $u$, let $\mathcal{M}$ be the set of monotone maps $\mathbb{R} \to \mathbb{R}$. The ceiling is

$$C_A(u) \;=\; \min_{m \in \mathcal{M}} \; \frac{1}{N} \sum_{i=1}^{N} \big| m(u_i) - H_i^* \big|, \tag{5}$$

computed by isotonic regression *fitted on the evaluation split*. It therefore has no generalisation gap and is optimistic by construction; we report the minimum of (5) over all candidate scalars. A monitor that beats it uses structure that no monotone function of any single uncertainty scalar can express.

# 4 World Models and Architectures

We train three world models that share data, seeds, targets and all downstream code, differing only in dynamics (Figure 1b, Table 1). M1 and M2 are GRU-based recurrent state-space models of different capacity. M3 replaces recurrence with causal self-attention and has *no autoregressive latent feedback*: observations are replaced by a learned mask token wherever unavailable, so imagination is a single masked forward pass rather than a rolled-forward prior. M3 exposes the same interface and produces the same head set, so $H^*$, the features, and every downstream computation are identical across architectures.

# 5 Experimental Setup

**Configurations.** A (`DoorKey-16x16`) is the training distribution. Shifts: B (`FourRooms`, unseen layout topology), C (A with 15% action slip), D (`MultiRoom-N6`, long corridors), E (`SimpleCrossingS11N5`, unseen maze family), F2 (observation noise), F3 (20% action dropout). World models and monitors are fitted on A unless stated.

**Protocol.** All decision rules were frozen before the corresponding evaluation across five pre-registration documents; departures are logged with timing. Layout seeds are disjoint across world-model training, monitor training, validation and test, and the monitor trains on episodes the world model never saw, so $H^*$ is not inflated by memorisation. Model selection uses validation only. We use three training seeds with $K = 5$ ensemble members, and report episode-clustered bootstrap intervals over 2000 resamples.

**Two measurement details that affect comparisons.** First, 21% of A-test targets sit at the ceiling $H_{\max} = 30$ and are therefore right-censored; MAE is inflated and the upper tail is unidentified. Second, the main run evaluates shifts on the *full* configuration split, whereas the transfer experiments (§6.5–§6.7) partition each shifted configuration 60/20/20 by episode and evaluate only on the held-out 20%. Baselines therefore differ slightly between sections — for example the no-exposure monitor reaches $\rho = 0.165$ on D in Table 4 and $\rho = 0.135$ on D's test portion in Table 5 — and numbers should be compared within a table, not across.

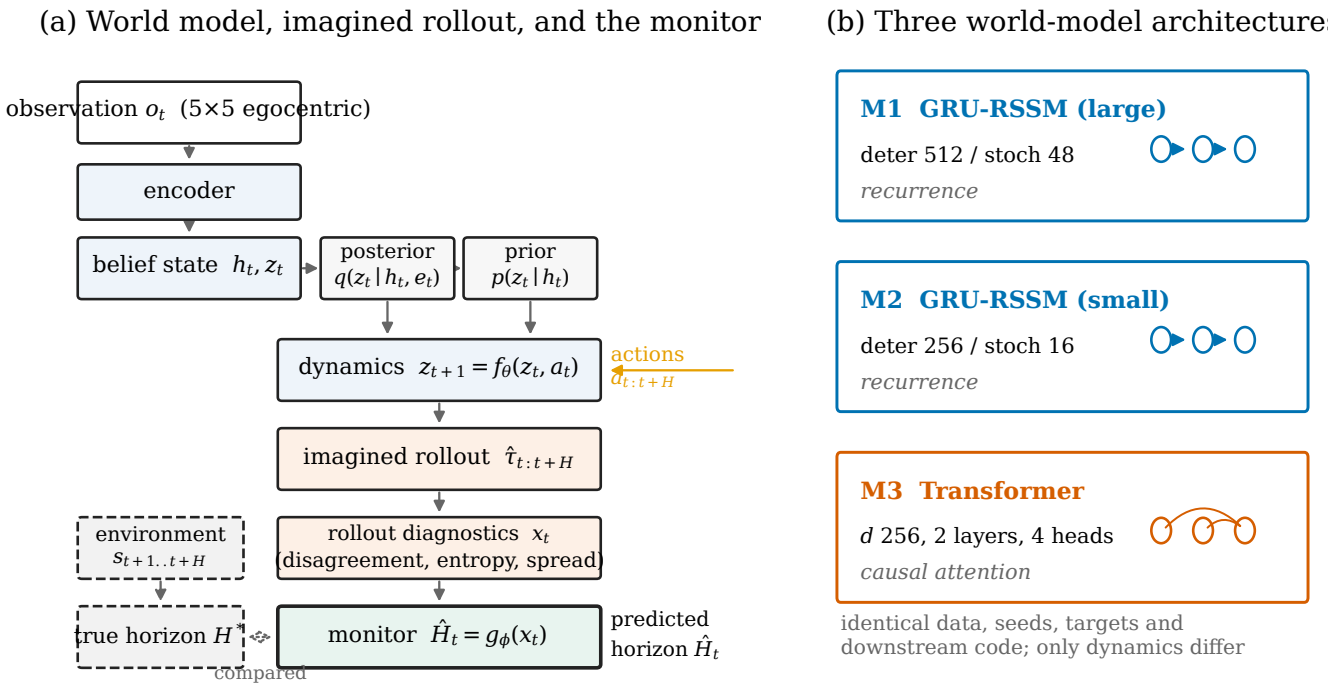



**Figure 1:** **(a)** The world model produces an imagined rollout from a belief state and a planned action sequence; rollout diagnostics $x_t$ feed the monitor $g_\phi$, whose output $\hat{H}_t$ is compared against the reliability horizon $H^*$ measured against the environment (dashed path, available only at evaluation time). **(b)** The three architectures. M1/M2 propagate a recurrent state; M3 aggregates history by causal attention and has no autoregressive latent feedback.

| | architecture | dynamics | size | $H^*$ on A | MAE ↓ | $\rho$ ↑ | vs. ceiling |
|---|---|---|---|---|---|---|---|
| M1 | GRU-RSSM (large) | gated recurrence | deter 512, stoch 48 | 15.59 | 6.408 | +0.584 | +1.355 |
| M2 | GRU-RSSM (small) | gated recurrence | deter 256, stoch 16 | 13.65 | 6.351 | +0.551 | +1.050 |
| M3 | Transformer | causal attention | $d$ 256, 2 layers, 4 heads | 16.58 | 6.432 | +0.513 | +0.993 |

**Table 1:** World-model architectures, with in-distribution monitor performance and the margin over that architecture's own Claim-A ceiling. Single seed, $K = 3$ ensemble members.

**Hypothesis ledger.** Table 2 lists all nine pre-registered hypotheses with their frozen verdicts. Five were not supported. We report them in the main text rather than the appendix.

# 6 Results

## 6.1 The monitor predicts the horizon

On A-test, $H^*$ has mean 15.60 and interquartile range 20–21, so the target is not degenerate. Table 3 reports the comparison and Figure 2 shows it per state.

The monitor attains MAE $6.416 \pm 0.087$ and $\rho = 0.573 \pm 0.011$ (mean ± s.d. over three seeds; the feature-group variant selected on validation differed by seed, so we also report fixed variants). Against the optimistic ceiling of (5) it improves by 1.080, 1.102 and 0.936 MAE on the three seeds, with paired bootstrap lower bounds 0.978, 1.010 and 0.854 — all excluding zero. Against a constant predictor it improves by 2.51–2.63 MAE. In distribution the predictive distribution is well calibrated, with expected calibration error 0.023–0.041 for $P(H^* \geq h)$, $h \in \{3, 5, 10, 20\}$.

Two comparisons sharpen the result. The monitor beats *online truncation* — stopping the rollout when imagined displacement spread exceeds one cell — by 1.99 MAE. And a *privileged* predictor given the true local $11 \times 11$ grid together with the ordered action sequence performs *worse* ($\rho = 0.226$) than the monitor. Knowing the environment is not sufficient; what the monitor exploits is the model's relationship to it.

## 6.2 Rollout information is necessary

Restricting $g_\phi$ to strictly time-$t$ features — belief state, one-step uncertainty, planned actions, statistics of the current observation, and no quantity computed from the imagined rollout — costs the effect entirely: MAE $8.065 \pm 0.087$, which is *worse* than the ceiling by 0.464, 0.630 and 0.735 on the three seeds. The signal the monitor exploits lives in the internal disagreement of the imagined trajectory. The honest reading of "before observing the future" is therefore: before observing it, but after

| | question | protocol | verdict | note |
|---|---|---|---|---|
| H1 | Uncertainty correlates with future error | main run | Supported | weak; the premise, not the result |
| H2 | Monitor beats best monotone single-scalar map | main run, 3 seeds | **Supported** | +1.04 MAE, CIs exclude 0 |
| H2′ | Same, without imagined-rollout features | main run, 3 seeds | Not supported | rollout information is necessary |
| H3 | Horizon drops before failure under shift | main run, 3 seeds | Not supported | fails anti-conservatively |
| H4 | Adaptive horizon improves planning at lower compute | 60 paired episodes | Not supported | *weak test*: monotone frontier |
| H5 | OOD failure repaired by rank-norm / novelty gate | 3 seeds, 4 arms | Not supported | rank-informativeness, not scale |
| H6 | Shift exposure restores transfer | 3 seeds + size-matched | **Supported** | $\rho$ 0.135 → 0.766 on unseen D |
| H7 | Phenomenon is architecture-general | 3 architectures, 1 seed | Supported (H2); partial (H6) | monitor only partly portable |
| H8 | Shift-type taxonomy predicts transfer | 3 seeds, 2 new configs | Not supported | geometric half confirmed; F invalid |
| H9 | Noise-driven shifts transfer to each other | 3 seeds, validity gate | Not supported | 0.444 vs 0.50 bar; taxonomy withdrawn |

**Table 2:** All pre-registered hypotheses and their frozen verdicts. H4 is labelled a weak test because the planning task's compute–performance frontier turned out to be monotone, leaving no interior optimum for an adaptive horizon to find. H8/H9 withdrew a proposed shift-type taxonomy; §6.7 reports the asymmetry the data support instead.

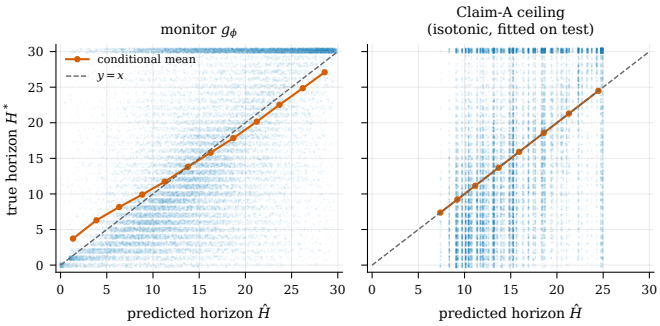



**Figure 2:** Predicted versus true reliability horizon, pooled over three seeds ($3 \times 11{,}100$ states). Orange: conditional mean in bins of the prediction. The optimistic ceiling (right) compresses its predictions into a narrow band and cannot express the extremes of the target.

| method | MAE ↓ | Spearman $\rho$ ↑ | $R^2$ ↑ |
|---|---|---|---|
| **Monitor** $g_\phi$ (per-seed selected) | **6.416** ± 0.087 | **0.573** ± 0.011 | **0.318** |
| `no_latent` | 6.394 ± 0.068 | 0.580 ± 0.005 | 0.331 |
| `full` | 6.457 ± 0.080 | 0.566 ± 0.005 | 0.304 |
| uncertainty only | 6.604 ± 0.109 | 0.553 ± 0.012 | 0.301 |
| Claim-A ceiling (isotonic, on test) | 7.456 ± 0.158 | 0.459 | 0.224 |
| Strict a-priori (no rollout) | 8.065 ± 0.087 | 0.372 ± 0.003 | 0.119 |
| Hand-crafted + actions | 8.274 ± 0.095 | 0.320 ± 0.005 | 0.107 |
| Online truncation | 8.384 ± 0.082 | 0.298 ± 0.011 | 0.096 |
| Privileged: true local grid | 8.628 ± 0.067 | 0.226 ± 0.006 | 0.030 |
| Fixed horizon (median of val) | 8.975 ± 0.101 | 0.000 ± 0.000 | -0.010 |
| Random horizon | 11.002 ± 0.047 | 0.001 ± 0.004 | -0.721 |

**Table 3:** In-distribution horizon prediction on A-test, mean ± s.d. over three seeds. The Claim-A ceiling is fitted on the evaluation split and is therefore optimistic. Bold marks the primary monitor only.

imagining it.

## 6.3 Under shift the monitor fails anti-conservatively

Table 4 and Figure 3 report the shift results. In distribution the monitor is nearly unbiased. Under shift the true horizon collapses to 4–8 while the prediction *rises* to 13–21, and the ranking quality falls from $\rho = 0.573$ to 0.165–0.271.

The direction matters. A monitor that degraded to noise would be useless; one that becomes *more* confident exactly where the world model is least reliable is worse than useless for any application that would gate imagination on it.

## 6.4 Simple repairs do not recover it

We tested two corrections, neither fitted on shifted data: replacing each uncertainty feature by its within-split quantile rank, and multiplying the prediction by a novelty gate $\exp(-\lambda \max(0, \nu - \tau))$ whose constants are set on in-distribution validation so that the gate is inert in distribution. Both fail on all three seeds (Table 4); predictions remain near 20 against true horizons of 4–8, and rank-normalisation makes configuration C worse. §7 explains why.

## 6.5 Exposure to a shifted distribution restores transfer

We refit $g_\phi$ on combinations of configurations, retraining no world model. Each shifted configuration is partitioned 60/20/20 by episode with the test portion evaluation-only in every regime, so "seen" and "unseen" cells are directly comparable (Table 5, Figure 4).

On D, never seen in training, the regime A+B+C moves MAE from 16.768 to 3.584 and $\rho$ from 0.135 to 0.766, and the prediction becomes appropriately conservative. A size-matched control, subsampling every regime to the A-only anchor count, reproduces the effect (MAE 3.622,

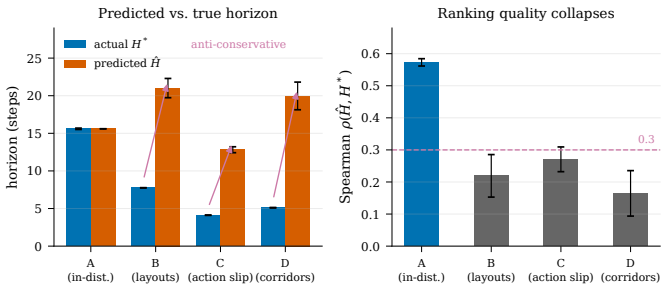



**Figure 3:** Left: true and predicted horizon by configuration; the gap opens in the unsafe direction under shift. Right: ranking quality against the 0.3 reference used in the frozen decision rules. Bars are means over three seeds, error bars s.d.

| configuration | true $H^*$ | predicted $\hat{H}$ | Spearman $\rho$ |
|---|---|---|---|
| A (in distribution) | 15.60 | 15.67 | +0.573 |
| B (unseen layouts) | 7.75 | **21.01** | +0.219 |
| C (action slip) | 4.11 | **12.82** | +0.271 |
| D (long corridors) | 5.11 | **19.97** | +0.165 |
| *Repair attempts (H5), predicted $\hat{H}$ on each split* | | | |
| no repair | B 21.56 | C 13.59 | D 21.15 |
| rank-normalised inputs | B 20.99 | C 16.08 | D 21.29 |
| novelty gate | B 20.25 | C 13.12 | D 19.95 |
| both | B 19.78 | C 15.56 | D 20.12 |

**Table 4:** Distribution-shift failure and the two repair attempts. Predicted horizons rise where true horizons fall. Neither repair restores conservatism. Evaluated on full configuration splits.

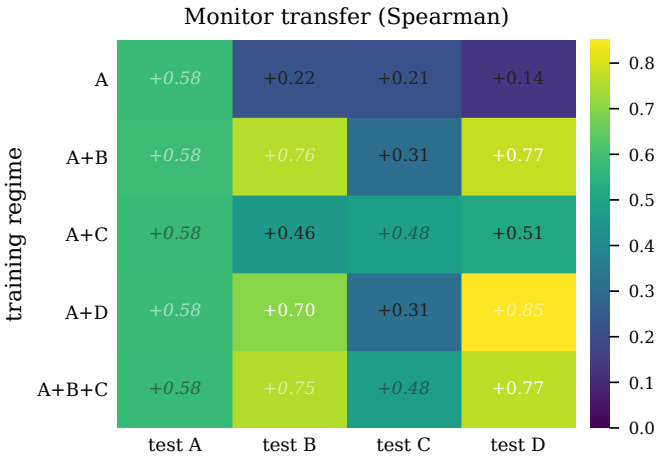



**Figure 4:** Monitor transfer. Rows are training regimes, columns held-out configurations; italic cells denote a configuration seen during monitor training. Exposure to one shift lifts performance on configurations never seen.

| training regime | test A | test B | test C | test D |
|---|---|---|---|---|
| A | *+0.580* | +0.218 | +0.213 | +0.135 |
| A+B | *+0.584* | *+0.755* | +0.306 | +0.771 |
| A+C | *+0.576* | +0.461 | *+0.479* | +0.510 |
| A+D | *+0.580* | +0.699 | +0.311 | *+0.847* |
| A+B+C | *+0.577* | *+0.754* | *+0.477* | +0.766 |

**Table 5:** Spearman $\rho$ by training regime, mean over three seeds. Italic entries are configurations seen in training. Evaluated on 20% held-out portions.

$\rho = 0.762$). We therefore conclude that the effect is *not explained by sample count alone*; we do not claim that diversity is its only cause. Adding a second shift changes little (A+B $\to$D gives $\rho = 0.771$), so within this environment family the marginal value of further shifts was small.

## 6.6 Architecture generality and monitor portability

All three architectures beat their own Claim-A ceiling on their own $H^*$: margins +1.355, +1.050 and +0.993 MAE, with paired lower bounds 1.251, 0.945 and 0.901 (Table 1). The core phenomenon is therefore not an artefact of recurrent latent feedback — it appears in a model that has none. **These runs use a single seed with** $K = 3$; they establish existence, not a precision estimate, and we make no claim of statistical equivalence between architectures.

Applying a monitor fitted on one architecture to another (Table 6, Figure 5) gives $\rho = 0.513$–$0.584$ within an architecture, 0.356–0.391 within the GRU family, and 0.207–0.362 across families. Transfer is clearly above zero, so part of the uncertainty–horizon relationship is architecture independent, but a monitor cannot be reused as-is.

The restoration effect of §6.5 *does not* replicate in the Transformer: $\rho$ on held-out D moves only from 0.257 to 0.290. The Transformer is a considerably better world model under shift ($H^* = 8.55$ on D against $\approx 5.1$ for the GRUs, with 3.8% of anchors collapsing to $H^* = 0$ against $\approx 29\%$). For the GRUs, much of what exposure teaches is "under shift, expect near-zero" — a large and easily learned correction. This is consistent with the restoration effect scaling with how badly the underlying model degrades, though we test a single contrast and cannot establish that relationship.

## 6.7 Restoration is selective across shift types

We pre-registered a two-family account of transfer and tested it predictively on two new configurations. The geometric half was confirmed: on E, never seen and drawn from a different maze family, A+B $\to$E reaches $\rho = 0.855$ against a no-exposure baseline of 0.116, and E acts as a donor (A+E $\to$D gives 0.800). The noise-driven half was not: A+C $\to$F3 reaches 0.444 against a pre-registered bar of 0.50. **We therefore withdraw the taxonomy** and report only the asymmetry the data support (Figure 6):

| donor $\downarrow$ / recipient $\to$ | geometric | noise-driven |
|---|---|---|
| geometric (B, E) | 0.73–0.86 | $\approx 0.31$ |
| noise-driven (C, F3) | 0.39–0.54 | 0.44–0.47 |

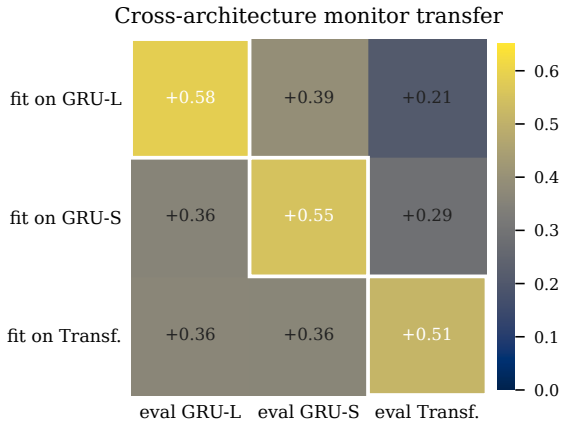



**Figure 5:** Cross-architecture monitor transfer (Spearman). Boxed cells are same-architecture references. Single seed, $K = 3$.

| | eval GRU-L | eval GRU-S | eval Transf. |
|---|---|---|---|
| fitted on GRU-L | *+0.584* | +0.391 | +0.207 |
| fitted on GRU-S | +0.356 | *+0.551* | +0.288 |
| fitted on Transf. | +0.363 | +0.362 | *+0.513* |

**Table 6:** Monitor fitted on one architecture, evaluated on another's A-test. Italic entries are same-architecture references. Single seed, $K = 3$.

Geometric donors transfer strongly to geometric recipients and weakly to noise-driven ones; noise donors transfer moderately to everything. §7 gives the mechanism.

Two caveats. The pattern rests on four donors within one environment family. And donors differ in *severity* as well as in kind, so an account in which transfer depends on how much the donor degrades the model — rather than on the type of shift — is not excluded by these experiments.

### 6.8 A weak test of downstream utility

We also tested whether the monitor improves planning by setting the imagination horizon adaptively in a model-predictive controller. It does not: adaptive planning reached success $0.517 \pm 0.065$ against $0.933 \pm 0.032$ for the best fixed horizon, at $5.49\times$ the imagination compute. However, success fell *monotonically* with horizon, so the compute–performance frontier had no interior optimum and there was nothing for an adaptive rule to find; our attempt to make the task non-myopic did not take (mean detour 0.13 cells). We report this as a **weak test** rather than a refutation. Details in the appendix.

## 7 Mechanistic Analysis

Why does the monitor fail under shift, and why do input corrections not help?

Write $\rho_D(u) = \rho\big(u, H^* \mid D\big)$ for the within-distribution rank correlation between a scalar feature $u$ and the horizon on distribution $D$. It is useful to separate two things

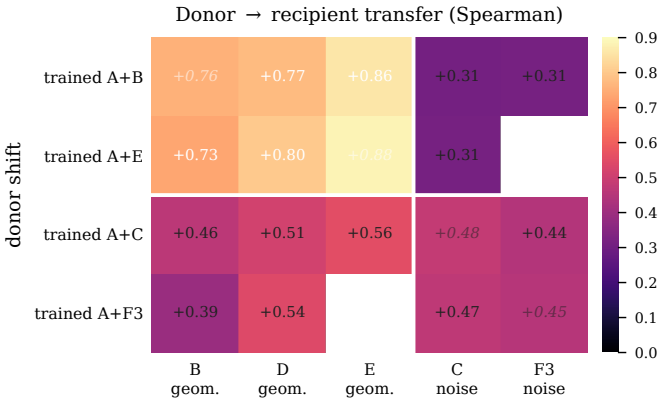



**Figure 6:** Donor–recipient transfer. White lines separate geometric from noise-driven configurations; italic cells evaluate a donor on itself. The top-right quadrant — geometric donors on noise-driven recipients — is the weakest.

a shift can do to a feature:

$$\text{scale shift:} \quad \mathbb{E}_{D'}[u] \neq \mathbb{E}_D[u], \quad \rho_{D'}(u) \approx \rho_D(u), \tag{6}$$

$$\text{rank shift:} \quad |\rho_{D'}(u)| \ll |\rho_D(u)|. \tag{7}$$

A monotone reparameterisation of $u$ — rank normalisation included — can absorb a scale shift. It cannot absorb a rank shift, because it preserves the ordering it is given, and under a rank shift that ordering has stopped carrying information about $H^*$.

The dominant in-distribution feature, the spread of imagined displacement across ensemble members, undergoes a rank shift. Its mean *rises* from 0.517 in distribution to 0.986, 1.455 and 0.971 on B, C and D — so aggregate separation between distributions is retained — while $\rho_D$ collapses from $-0.446$ to $-0.111$, $-0.191$ and $-0.089$. The signal still reports *that* matters are harder; it stops reporting *which states* will fail. This is the direct explanation of §6.4: rank normalisation is exactly the operation that cannot help here.

Two further observations. Predictive entropy moves in the *wrong* direction under shift, its $\rho_D$ going from $-0.394$ in distribution to $+0.001$ on D, whereas ensemble spread retains its sign. Berger et al. [2026] report a related but distinct effect: uncertainty *declining along* an extended latent rollout as it is drawn into well-modelled attractor regions. We measure across-distribution means and within-distribution rank correlations rather than the within-rollout trajectory of uncertainty, so our observations neither confirm nor contradict that account; establishing whether the two mechanisms coincide would require measuring uncertainty along the rollout, which we did not do. What our data do show is that the failure is not confined to one signal family: entropy and ensemble spread degrade differently, and no feature we measured is both stable and useful — the feature with the lowest rank instability has $\rho \approx 0$ everywhere (Figure 7).

That figure also explains the asymmetry of §6.7. One feature — how far the agent can currently see unobstructed — is *useless* in distribution ($\rho = +0.004$) and

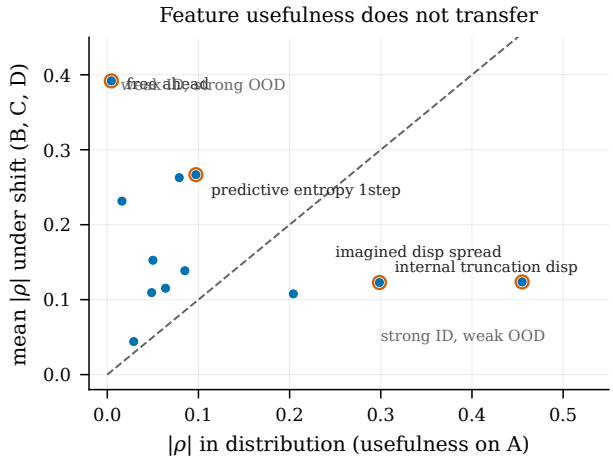



**Figure 7:** In-distribution usefulness against usefulness under shift, per uncertainty feature. Points far below the diagonal undergo a rank shift; the circled point near the $y$-axis is useless in distribution yet dominant under geometric shift.

*dominant* under geometric shift (+0.586, +0.566, +0.533 on B, D, E). A monitor trained only on A has no reason to use it; exposure to a geometric shift teaches it to. On noise-driven configurations the same feature is inert (+0.024, −0.015, −0.031). A cue that rescues geometric shifts therefore cannot rescue noise-driven ones, and we find no equivalent single carrier on the noise side — which is why those donors transfer broadly but never strongly.

# 8 Limitations

**One environment family.** All experiments use MiniGrid with symbolic observations and world models of roughly 2.8M parameters. The mechanism we identify — a cue that is useless in distribution and dominant under shift — is instantiated by a domain-specific feature, and whether an analogue exists for pixel-based or continuous-control world models is untested.

**Censoring.** 21% of A-test targets sit at $H_{\max} = 30$. MAE is inflated and the upper tail of $H^*$ is unidentified; a larger $H_{\max}$ would settle this.

**Effect size.** $R^2 = 0.318$: most variance in $H^*$ is unexplained, and the margin over the optimistic ceiling is about 1 MAE against a target with s.d. $\approx 10.3$. The result is statistically robust and quantitatively modest.

**Architecture evidence is single-seed.** §6.6 uses one seed and $K = 3$ per architecture. It establishes that the phenomenon occurs without recurrence; it does not support comparisons between architectures.

**Few shift families, and a severity confound.** The asymmetry rests on four donors. Two configurations (observation noise at 15% and 70%) failed a pre-registered validity gate because they barely degraded the model, so the noise-driven side is supported by two instruments rather than three. Donor severity and donor type covary.

**Downstream utility untested.** The planning experiment was a weak test (§6.8); we make no claim that the monitor improves control.

# 9 Discussion

Our results support a bounded statement: a world model can learn a useful monitor of the reliability horizon of its imagined futures, but that monitor is distribution-dependent and only partially transferable across architectures and shift types.

The practical reading is uncomfortable for uncertainty-gated imagination. A monitor fitted where the model works will be most confident where the model fails, and the two obvious input-level corrections do not help, because the problem is that features stop *ranking* states rather than that they are mis-scaled. What does help is exposure to failures — but selectively, and less so when the world model is already strong under shift.

This suggests reading reliability monitoring less as reading out an intrinsic property of uncertainty and more as a learned model of one's own failures, whose validity is inherited from the distribution of failures encountered during training. The asymmetry sharpens the point: what a monitor learns from one kind of failure may be a specific cue rather than a general capacity for caution.

# 10 Conclusion

We defined an operational reliability horizon for imagined rollouts and showed that a learned monitor predicts it beyond an optimistically fitted monotone single-scalar ceiling, provided it may inspect the imagined rollout. The monitor fails anti-conservatively under distribution shift; rank normalisation and novelty gating do not repair it; exposure to an appropriate shifted distribution substantially restores transfer without being explained by sample count alone; and that restoration is selective across shift types and does not replicate in a world model that degrades less. Of nine pre-registered hypotheses, three were supported and five were not; the negative results carry most of the content.

# Reproducibility

All decision rules, thresholds, splits and seeds were frozen before the corresponding evaluations in five pre-registration documents, with departures logged. Every number in this paper is emitted programmatically from stored result artifacts. Code, protocols, the deviation ledger and per-seed statistics accompany the submission; the appendix documents architectures, hyperparameters, feature definitions, the bootstrap procedure, censoring, and the two configurations rejected as invalid instruments.

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